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Institut Teknologi Nasional

ETS
IFB-451 BIG DATA

MID-TERM GROUP ASSIGNMENT

5-Minute Big Data Discovery and Demonstration Video

A Healthcare Informatics Approach Using the 5Vs of Big Data



Prepared by:

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Academic Year: 2026/2027

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Department: Informatics Department, Institut Teknologi Nasional (Itenas) Bandung

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1. Assignment Overview

Component	Description
Course	IFB-451 BIG DATA
Assessment	Mid-Term Group Assignment (ETS)
Assessment Weight	35%
Group Size	5 students
Learning Modules	Modules 1–7
Student Competencies	SC1, SC2 and SC3
Presentation Format	5-minute YouTube video
Presentation Approach	Project & Presentation (P&P)

Component	Description
Primary Focus	Practical demonstration of the 5Vs of Big Data
Application Domain	Healthcare and Public Health Informatics
Technical Requirement	Functional Big Data pipeline prototype
Deliverables	Video demonstration and three-page PDF
Submission Platform	Itenas eLearning
Submission Deadline	10 November 2026

2. Purpose

This assignment examines the application of the five characteristics of Big Data—**Volume, Velocity, Variety, Veracity and Value**—to healthcare information processing. Working in groups of five, students will investigate a healthcare problem and develop a functional prototype using the concepts and technologies introduced in Modules 1–7.

Healthcare datasets present challenges beyond the quantity of information they contain. Clinical records may involve complex relationships, repeated observations, different data representations and demanding processing requirements. Students are expected to examine these characteristics, justify their technical approach and demonstrate how the resulting information contributes to understanding the selected healthcare problem.

Central Assignment Question

How do the 5Vs influence the design and implementation of a healthcare data-processing pipeline, and what meaningful healthcare information can be obtained from it?

The five-minute presentation must address this question through evidence from the group's working prototype.

3. The 5Vs: Core Demonstration Requirements

The 5Vs provide the basis for examining the selected dataset and evaluating the proposed technical solution. Each characteristic must be demonstrated through the group's implementation.

Table 3.1. The 5Vs: Conceptual Understanding and Technical Demonstration

Characteristic	Conceptual Understanding	Healthcare Context	Required Demonstration	Expected Evidence
Volume	Quantity of data generated, stored and processed. Structural relationships and computational demands affect processing but are not themselves measures of Volume.	Longitudinal EHRs contain encounters, diagnoses, medications and laboratory observations that may require complex relational processing.	Examine data quantity, structure and relationships; demonstrate aggregation or relational processing.	Record counts, structure, processing operations and measured performance.
Velocity	Rate of data generation and arrival, considered alongside required processing timeliness.	Monitoring systems may produce observations every few seconds; admission records may be updated periodically.	Demonstrate periodic or simulated continuous arrivals; measure ingestion frequency and processing latency.	Arrival timestamps, update frequency and latency.
Variety	Differences in data sources, formats, structures and representations.	Clinical encounters, laboratory results and monitoring observations may use different schemas.	Process and integrate at least two formats or representations.	Source schemas, transformations and integrated records.
Veracity	Accuracy, completeness, consistency and reliability of data.	Records may contain missing values, duplicates, inconsistent units or incorrect timestamps.	Identify at least two quality problems and apply appropriate validation or correction procedures.	Before-and-after quality measures and documented handling decisions.
Value	Usefulness of the information produced through processing and analysis.	Admission patterns may inform capacity planning; surveillance records may reveal changes in reported incidence.	Answer a defined healthcare question and explain the results and their limitations.	Query results, visualisation and healthcare interpretation.

Table 3.2. Minimum Demonstration Requirements and Assessment Considerations

Characteristic	Acceptable Evidence	Insufficient Evidence
Volume	Demonstrated data scale, relationships and processing requirements.	File size or record count without processing analysis.
Velocity	Measured arrival frequency, ingestion and processing latency.	Query execution time alone.
Variety	Integration of two different formats or representations.	A list of formats without integration.
Veracity	Identified quality issues, justified procedures and measured results.	An unsupported claim that data was cleaned.
Value	Findings interpreted in relation to the healthcare problem.	Charts without interpretation.

The selected dataset must provide sufficient scope to investigate all five characteristics. A large dataset is not compulsory, but students must explain the technical challenges presented by their data and demonstrate how these are addressed.

The assessment considers the appropriateness of the technical approach and the evidence presented, not the size of the dataset or complexity of the software alone.

4. Learning Outcomes and Module Alignment

The assessment covers Modules 1–7 and their corresponding Student Competencies.

SC	Modules	Learning Focus	Expected Demonstration
SC1	M1–M2	Big Data concepts, characteristics, ecosystem and architecture.	Analyse the 5Vs and justify the selected processing architecture.
SC2	M3–M4	Data storage, management and Hadoop ecosystem.	Demonstrate data storage, integration, structural complexity and quality management.

SC	Modules	Learning Focus	Expected Demonstration
SC3	M5–M7	MapReduce, Spark, Spark SQL and DataFrames.	Implement data processing operations, evaluate performance and produce analytical findings.

The modules should be presented as connected components of a single data processing workflow.

5. Healthcare Case Study Selection

Each group must select **one healthcare or public health case study** and use it consistently throughout the five-minute demonstration.

The following cases provide opportunities to examine different data structures, processing requirements and healthcare applications. All five characteristics must be addressed, regardless of the selected topic.

Table 5.1. Healthcare Case Studies Mapped to the 5Vs

Case Study	Volume	Velocity	Variety	Veracity	Value
Longitudinal Electronic Health Record Analytics	Encounters and linked clinical observations.	Periodic encounter updates.	Encounter and JSON observation records.	Duplicates and missing observations.	Healthcare utilisation patterns.
Hospital Resource Utilisation	Admissions and occupancy calculations.	Admission and discharge updates.	Admissions and bed-capacity records.	Missing times and inconsistent occupancy.	Demand and capacity planning.
COVID-19 Epidemiological Analytics	Notifications by region and period.	Daily or weekly reporting.	Cases, vaccination and population data.	Delays, duplicates and missing regions.	Reported infection trends.
Clinical Laboratory Data Analytics	Repeated tests and related encounters.	Incoming result batches.	Laboratory and encounter records.	Missing results and inconsistent units.	Laboratory utilisation.
Population Health Surveillance	Disease reports across regions and groups.	Periodic surveillance updates.	Reports, demographics and geography.	Incomplete reports and inconsistent dates.	Regional disease patterns.
Healthcare Insurance Claims Analytics	Claims linked to providers and services.	Periodic claims submissions.	Claims, providers and service records.	Duplicate claims and billing inconsistencies.	Unusual claims patterns.

Case Study	Volume	Velocity	Variety	Veracity	Value
IoT-Based Patient Monitoring	Repeated physiological time-series observations.	Simulated timed observations.	Heart rate, SpO_2 and blood pressure.	Missing or implausible readings.	Monitoring trends.
Chronic Disease Population Analytics	Longitudinal diabetes or hypertension records.	Periodic follow-up updates.	Measurements, demographics and encounters.	Missing values, duplicates and unit issues.	Follow-up and disease patterns.

5.2 Case Study Selection Requirements

The selected case study must support the complete technical demonstration described in Section 3 and the five-minute video structure.

Students may use public, synthetic or appropriately de-identified healthcare datasets. Where necessary, simulated updates may be introduced to demonstrate Velocity.

Alternative healthcare topics are acceptable provided they address all five Vs and produce an interpretable analytical result.

6. Tools and Technical Environment

Students may select technologies appropriate to their case study and the competencies assessed in this assignment.

Table 6.1. Suggested Tools and Technologies

Category	Suggested Technology	Application
Distributed Storage	Hadoop HDFS	Storage and management of distributed datasets.
Data Processing	Apache Spark / PySpark	Data transformations, aggregations and processing.
Query Processing	Spark SQL / Hive	Analytical queries and relational operations.
Programming	Python	Data preparation and processing.
Data Formats	CSV, JSON, Parquet	Representation and integration of heterogeneous data.

Category	Suggested Technology	Application
Data Quality	pandas / PySpark	Validation and cleaning procedures.
Visualisation	Plotly / Power BI / Grafana	Presentation of analytical results.
Development Environment	Jupyter Notebook / Google Colab / Local Spark	Prototype development and demonstration.

Technical Scope

Students are not required to implement every technology listed above.

A functioning PySpark-based prototype is acceptable where it supports the required demonstrations.

Students should select technologies according to the characteristics of their data and the processing operations required.

Where computational resources are limited, students may use synthetic datasets, scaled experiments or simulated data arrival.

Simulated conditions must be identified clearly.

7. Five-Minute Video Structure and Student Roles

The presentation must involve all five group members and demonstrate a single healthcare data processing workflow.

Table 7.1. Video Structure and Student Responsibilities

Segment	Module(s)	Student Role	Duration	5Vs Focus	Required Demonstration
1. Healthcare Problem and Data Characteristics	M1	Project Lead / Analyst	0:45	All 5Vs	Introduce the healthcare problem, dataset and relevant characteristics, including data scale and complexity.

Segment	Module(s)	Student Role	Duration	5Vs Focus	Required Demonstration
2. Big Data Architecture and Data Flow	M2	System Engineer	0:45	Volume and Velocity	Present the architecture and explain how it addresses data structure, processing requirements and data arrival.
3. Data Storage, Integration and Quality	M3–M4	Data Engineer	1:00	Variety and Veracity	Demonstrate data formats, integration, structural relationships and data validation.
4. Data Processing and Computational Requirements	M5–M6	Big Data Developer	1:15	Volume and Velocity	Execute processing operations, demonstrate computational complexity where relevant and report processing performance.
5. Analytical Results and Healthcare Interpretation	M7	Data Analyst	1:15	Veracity and Value	Execute analytical queries, present findings and explain their healthcare relevance and limitations.

Segment	Module(s)	Student Role	Duration	5Vs Focus	Required Demonstration
TOTAL	M1–M7	5 students	5:00	All 5Vs	Integrated technical demonstration

Table 7.2. Relationship Between the 5Vs, Teaching Modules and Student Responsibilities

5Vs	Relevant Modules	Student Responsibility	Video Segment
Volume	M1, M2, M5, M6	Examine data scale and complexity, justify processing requirements and demonstrate relevant operations.	Segments 1, 2 and 4
Velocity	M1, M2, M5, M6	Explain data arrival characteristics and demonstrate ingestion or processing timeliness.	Segments 2 and 4
Variety	M3, M4	Demonstrate heterogeneous data formats, schemas and integration procedures.	Segment 3
Veracity	M3, M4, M7	Identify data quality problems, apply validation procedures and assess their implications for analysis.	Segments 3 and 5
Value	M7	Execute analytical queries, interpret findings and establish their relevance to the healthcare problem.	Segment 5

7.3 Presentation Requirements

The video must contain evidence of a working prototype.

Students should demonstrate relevant code execution, data transformations, processing outputs and analytical results.

The five presentation segments must form one coherent demonstration.

Each student must contribute to the technical explanation according to their assigned role.

The presentation must not exceed five minutes.

8. Integrated Healthcare Demonstration Example

Case Study: Longitudinal Electronic Health Record Analytics

A healthcare organisation wants to analyse patient encounters and clinical observations to understand patterns of healthcare utilisation.

The dataset contains synthetic patient records, diagnoses, medications, laboratory observations and encounter histories.

Although the dataset may be relatively small, its structure contains relationships that require multiple transformations and analytical operations.

Table 8.1. Demonstration of the 5Vs

5V	Demonstration	Expected Evidence
Volume	Process patient encounters and related clinical observations using multiple joins and transformations.	Record counts, dataset relationships, query operations and processing measurements.
Velocity	Simulate periodic updates to clinical encounter records.	Update frequency, ingestion process and processing latency.
Variety	Integrate CSV patient encounter records with JSON clinical observations.	Source schemas, transformations and integrated output.
Veracity	Identify missing observations and duplicate encounters.	Validation results and data quality statistics.
Value	Analyse healthcare utilisation across encounters and reporting periods.	Analytical results and interpretation of service utilisation patterns.

8.2 Demonstrating Computational Complexity

Students may compare two processing operations.

Operation A — Simple Aggregation

Count patient encounters by department.

Operation B — Relational and Temporal Analysis

Integrate patient encounters, laboratory observations and diagnoses to examine clinical activity over time.

Table 8.2. Comparison of Processing Operations

Aspect	Operation A: Simple Aggregation	Operation B: Relational and Temporal Analysis
Data Structure	Single structured dataset.	Multiple interconnected datasets.
Processing Operations	Grouping and counting.	Joins, transformations and temporal analysis.
Data Relationships	Limited.	Multiple clinical and temporal relationships.
Computational Requirements	Primarily dependent on record count and aggregation operations.	Influenced by joins, transformations, data distribution and temporal processing.
Expected Demonstration	Encounter counts by department.	Integrated analysis of encounters, diagnoses and laboratory observations.

The comparison should identify differences in processing operations, execution time and computational requirements.

Students should explain whether the observed differences arise from data size, structural complexity, query design or other technical factors.

9. Submission Requirements

Each group must submit a five-minute YouTube video and a three-page PDF.

9.1 Video

The video must include:

1. Contributions from all five group members.
2. A clearly defined healthcare case study.
3. A functional data processing prototype.
4. Technical evidence addressing all five Vs.
5. Analytical findings relevant to the selected healthcare problem.
6. English or Bahasa Indonesia closed captions.

9.2 Three-Page PDF

Page	Required Content
Page 1 — Project Information	Case study title, group members, student numbers, dataset description and source.
Page 2 — Video and 5Vs Summary	YouTube link and a concise explanation of the evidence presented for each characteristic.
Page 3 — System Architecture	Diagram showing data sources, ingestion, storage, processing, integration, analysis and outputs.

Submission Platform

Itenas eLearning — IFB-451 BIG DATA

Submission Deadline

10 November 2026

File Naming Convention

GroupNumber_FullName_NIM.pdf

Each student must submit the group PDF individually using their respective name and student number.

10. Assessment Rubric

The assessment evaluates conceptual understanding, technical implementation and the interpretation of healthcare information.

Table 10.1. Assessment Criteria

Assessment Criterion	SC	Assessment Focus	Weight
Volume	SC1–SC3	Understanding of data scale, structural complexity and processing requirements.	**10%**

Assessment Criterion	SC	Assessment Focus	Weight
Velocity	SC1–SC3	Understanding of data arrival, ingestion and processing timeliness.	**10%**
Variety	SC1–SC2	Integration and processing of different data formats and representations.	**10%**
Veracity	SC2–SC3	Identification, validation and management of data quality problems.	**10%**
Value	SC3	Analytical findings and their relevance to healthcare understanding or decision-making.	**15%**
Big Data Architecture and Technical Implementation	SC1–SC3	Appropriateness and functionality of the processing architecture and prototype.	**25%**
Healthcare Case Study and Interpretation	SC1–SC3	Relevance of the selected problem and quality of analytical interpretation.	**10%**
Teamwork and Presentation	All	Balanced participation, clarity and adherence to the five-minute limit.	**10%**
TOTAL			**100%**

Table 10.2. Assessment Distribution

Component	Weight
Demonstration of the 5Vs	55%
Technical Implementation	25%
Healthcare Relevance	10%
Teamwork and Presentation	10%
TOTAL	**100%**

11. Expected Learning Outcomes

Upon completing the assignment, students should be able to:

Learning Outcome	Competency	Expected Achievement
LO1 — Big Data Understanding	Conceptual Understanding	Explain the five characteristics of Big Data and their implications for healthcare information processing.
LO2 — Data Characteristics and Complexity	Data Analysis	Examine healthcare datasets in terms of scale, structure, relationships, processing requirements and data quality.
LO3 — Technical Implementation	Applied Computing	Develop a functional data processing prototype using technologies appropriate to the selected healthcare problem.
LO4 — Data Integration and Quality	Data Management	Integrate different data representations and apply appropriate data validation procedures.
LO5 — Analytical Interpretation	Healthcare Informatics	Produce meaningful analytical findings and explain their potential contribution to healthcare understanding or decision-making.
LO6 — Technical Communication	Scientific Communication	Present the methodology, implementation and findings through a concise, evidence-based technical demonstration.

12. Academic Requirements

Students must demonstrate an understanding of the technical decisions underlying their implementation.

Table 12.1. Academic Requirements

Requirement	Description
Data Characteristics	Examine all five Vs in relation to the selected healthcare dataset.

Requirement	Description
Technical Justification	Select processing technologies and architecture appropriate to the data and analytical requirements.
Computational Considerations	Distinguish data size, structural complexity and processing demands where relevant.
Data Quality	Support validation and correction procedures with measurable evidence.
Healthcare Interpretation	Relate analytical findings to the selected healthcare problem and acknowledge relevant limitations.
Data Protection	Use public, synthetic or appropriately de-identified datasets. Identifiable patient information must not be disclosed.
Academic Integrity	Present work that the group can explain, reproduce and defend. External code, datasets and supporting materials must be appropriately acknowledged.

The assessment does not reward system complexity for its own sake.

Students will be evaluated on their understanding of the selected problem, the appropriateness of their methods and the evidence presented.